



IF-USOD: Multimodal information fusion interactive feature enhancement architecture for underwater salient object detection

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ABSTRACT

Underwater salient object detection (USOD) has garnered increasing attention due to its superior performance in various underwater visual tasks. Despite the growing interest, research on USOD remains in its nascent stages, with existing methods often struggling to capture long-range contextual features of salient objects. Additionally, these methods frequently overlook the complementary nature of multimodal information. The multimodal information fusion can render previously indiscernible objects more detectable, as capturing complementary features from diverse source images enables a more accurate depiction of objects. In this work, we explore an innovative approach that integrates RGB and depth information, coupled with interactive feature enhancement, to advance the detection of underwater salient objects. Our method first leverages the strengths of both transformer and convolutional neural network architectures to extract features from source images. Here, we employ a two-stage training strategy designed to optimize feature fusion. Subsequently, we utilize self-attention and cross-attention mechanisms to model the correlations among the extracted features, thereby amplifying the relevant features. Finally, to fully exploit features across different network layers, we introduce a cross-scale learning strategy to facilitate multi-scale feature fusion, which improves the detection accuracy of underwater salient objects by generating both coarse and fine salient predictions. Extensive experimental evaluations demonstrate the state-of-the-art model performance of our proposed method.

1. Introduction

Salient object detection (SOD) [1] is a fundamental problem in the field of computer vision [2], aimed at rapidly and accurately identifying prominent objects within a scene, thereby facilitating subsequent visual tasks [3] such as image segmentation, object detection, and style transfer [4]. Numerous classical SOD models have been proposed and implemented over the years. With the growing demand for underwater exploration, underwater salient object detection (USOD) has increasingly become a hot research topic [5]. Although several powerful SOD models have demonstrated competitive performance, the single-modal underwater salient object detection models often suffer from low localization accuracy due to challenges such as optical distortion and water pattern variations [6]. To address this issue, multimodal sensors have emerged [7], such as depth sensors [8]. Depth sensors [9], by measuring the distance between objects and the sensor, provide precise spatial perception capabilities [10,11], allowing depth images to clearly reveal the structure and contours of objects. In contrast, RGB sensors capture reflected light information, which includes rich texture and detail [12]. Therefore, the integration of depth and RGB images to

achieve complementary feature representation is highly promising for achieving more comprehensive underwater object detection.

Current popular multimodal fusion methods excel in highlighting salient objects and enriching texture details [13]. Typically, these methods adopt a dual-branch architecture [14] to extract RGB and X-modality features separately [15]. These features are then fused proportionally based on the texture features of the source image, the annotation masks of the objects, and spatial-channel guidance techniques [16]. However, classical multimodal fusion methods currently face the following challenges: (1) These methods often prioritize achieving better visual fusion, while paying little attention to their adaptability in real-world applications and downstream visual tasks [17]. (2) Existing multimodal fusion techniques struggle to accurately convey semantic information to higher-level downstream tasks [18,19], leading to a significant drop in performance for such tasks, as shown in Fig. 1(a). (3) Some vision methods are more susceptible to higher-level semantic information [20,21]. It should be noted that simply embedding semantic probability maps as conditions in specific layers is insufficient to support effective information interaction between different modalities [22].

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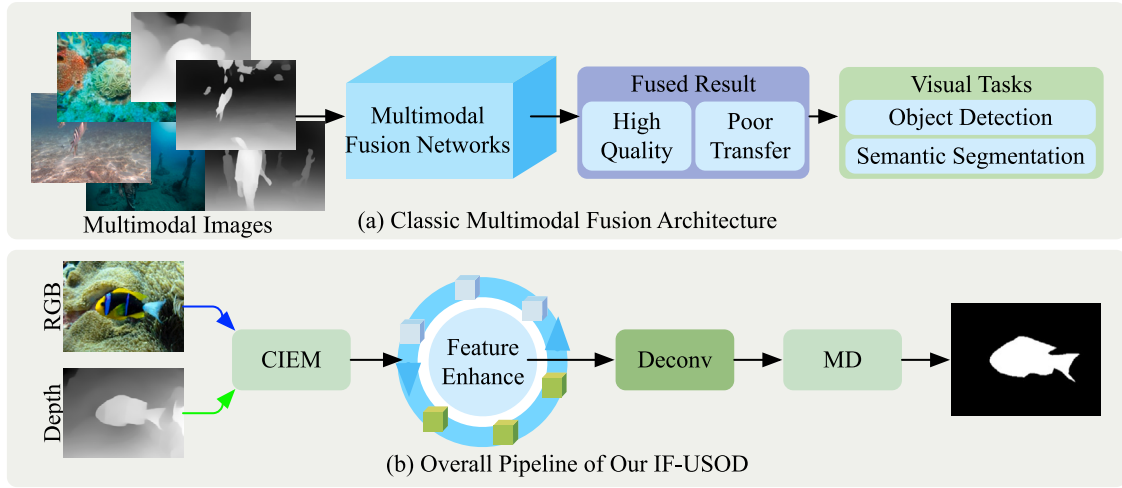


Fig. 1. Comparison between existing fusion methods and the proposed Interactive Feature Enhancement Architecture (IF-USOD). (a) Classic Multimodal Fusion Architecture. (b) Overall Pipeline of Our IF-USOD. Building on the classical multimodal fusion architecture, we first extract features using a cross-scale interactive embedding module (CIEM), followed by a re-weighted fusion of multimodal features to enhance cross-modal complementarity. Finally, we designed a multi-scale multimodal decoder (MD) to perform feature decoding across multiple scale levels.

Based on the aforementioned challenges, we propose to design a unified paradigm that integrates multimodal feature fusion with downstream tasks. Depth-visible underwater salient object detection requires the extraction of complementary features from two source images to achieve multimodal feature fusion and complete downstream visual tasks [23]. To further explore the collaborative relationships between multimodal features, we need to overcome three primary challenges: (1) Constructing a multimodal fusion network to capture, transmit, and fuse relevant multimodal semantic features, while simultaneously achieving complementary feature fusion at various levels and adapting to the underwater salient object detection task. (2) Developing an efficient bridging module to effectively transfer multimodal feature information to downstream tasks, thereby enabling image fusion to better enhance the performance of these tasks. (3) Designing a cross-scale collaborative learning strategy that allows the model to fully learn and utilize features from different modalities, thereby supporting more precise underwater object detection through multi-scale information integration.

In this work, we propose an Interactive Feature Enhancement Architecture for Underwater Salient Object Detection, termed IF-USOD, to learn fused features from depth and visible light images for underwater salient object detection, as illustrated in Fig. 1(b). Specifically, for the multimodal feature extraction component, considering that convolutional neural networks (CNNs) are effective at extracting local detail features from images while transformers excel at capturing long-range dependencies between features, we first construct a Cross-scale Interactive Embedding Module (CIEM) based on CNN and Transformer architectures. This module leverages the complementary strengths of CNNs and Transformers to better select and preserve the salient features of underwater objects. Next, to ensure that the fused features are effectively transmitted to downstream visual tasks, we design a Full-perception Cross-Attention module (FCA) to reweight the fused features, facilitating cross-modal complementarity and thereby ensuring the fused images enhance downstream visual tasks. Finally, we introduce a Multi-scale multimodal Decoder (MD), which reconstructs the underwater salient detection results by integrating multi-scale and multi-source features generated at different levels through an interactive transmission scheme. In the underwater salient object detection task, we supervise the detection results with accurately annotated saliency maps and construct a semantic loss function, which is then backpropagated through IF-USOD to encourage the network to generate fused features rich in semantic information. Our main contributions are as follows:

(1) We developed an interactive feature enhancement architecture for multimodal fusion of depth and visible light images in underwater salient object detection, exploring the feature bridging between multimodal fusion features and downstream visual tasks.

(2) By combining the strengths of CNNs and Transformers, we significantly highlighted salient features suitable for USOD tasks, while also guiding the fusion and propagation of multimodal features based on a cross-attention mechanism.

(3) Leveraging a cross-scale collaborative learning strategy, we effectively utilized fused features across different scales and levels, achieving more precise underwater salient object detection.

Experimental results demonstrate that the proposed IF-USOD fully integrates features across different modalities and exhibits outstanding performance in underwater salient object detection tasks.

2. Related works

Multimodal information fusion and multimodal salient object detection [24] are the central focuses of this work. In this section, we thoroughly review the existing techniques and trace the evolution of their development.

2.1. Multimodal information fusion

As pioneers in the field, researchers have utilized encoder-decoder architectures to achieve the extraction and fusion of multimodal features. Li and Wu [25] implemented feature extraction based on an encoder-decoder architecture, combining shallow and deep features to extract richer information from source images. To mitigate the loss of detail caused by continuous downsampling in the encoder-decoder architecture, Li et al. [26] introduced residual connections into the encoder-decoder framework to alleviate the issue of information loss. Zhou et al. [27] adopted a strategy of progressive guided fusion and depth enhancement to iteratively fuse and enhance multimodal feature information at each layer.

Multimodal information fusion methods based on deep CNNs primarily focus on network architecture and fusion strategies [28,29]. Xu et al. [30] followed a dense network architecture, integrating dense networks with information measurement within a single framework to adaptively estimate the importance of different modality features. To further address the poor performance of dense networks in capturing object structures and details, Liu et al. [31] employed an edge-detail attention-guided mechanism to enhance the network's ability to

highlight object structures and better capture rich detail information. They also adopted a coarse-to-fine network architecture to capture multi-scale information, thereby improving the network's feature representation capability. Subsequently, Tian et al. [32] introduced a coordinated weight allocation strategy that adaptively assigns fusion weights to better leverage the complementary information from multimodal features. Wu et al. [12] discovered a strong correlation between multi-granularity geometric priors and the hierarchical structure of neural networks, proposing a layered depth-aware strategy to achieve multi-level fusion of multimodal features. Considering the potential inflexibility of network architectures when dealing with different types of image data, Zhao et al. [33] proposed a correlation-driven feature decomposition fusion network to maintain the advantages of different modalities, making the fusion process more flexible and effective.

GAN-based Multimodal information fusion methods aim to construct feature constraints from a probabilistic distribution perspective, thereby promoting fused features with clearer and richer textures. Ma et al. [34] modeled the image fusion task in an adversarial manner between the generator and discriminator. However, using a single discriminator can lead to the generator treating features from different modalities equally, resulting in overly smooth fused images. Zhang et al. [35] utilized Transformers to optimize the generative adversarial network, where adversarial training aims to preserve the object structure and texture details, while the Transformer emphasizes the object structure of the fused image and enriches detail features, enabling the generator to focus on both local and global contexts. Le et al. [36] developed a unified image fusion architecture based on continual learning generative adversarial networks, employing elastic weight consolidation to prevent forgetting previously learned tasks.

Owing to the global receptive field and the ability to model long-range dependencies between adjacent pixels, Transformer-based image fusion methods have demonstrated superior performance [37]. Ma et al. [23] designed a general multimodal image fusion architecture by introducing an attention-guided cross-domain long-distance learning module, which performs well in structure preservation and detail retention. Zhang et al. [38] developed a convolution-guided Transformer for feature fusion, where the convolutional feature extractor is designed to learn local features, while the Transformer feature extractor captures long-range interdependencies among features. Subsequently, Liu et al. [39] employed Swin Transformer to extract hierarchical features and utilized attention mechanisms and edge information to optimize the fusion of features and salient boundaries between different modalities.

Existing multimodal information fusion methods exhibit limitations in detail capture, feature transmission, and cross-modal alignment, such as detail loss due to the encoder-decoder architecture, poor adaptability of dense networks, and overly smooth images generated by GANs. In contrast, the IF-USOD method enhances feature fusion and detail preservation through the combination of cross-scale interactive embedding of CNNs and Transformers, along with a fully-aware cross-attention mechanism. This effectively overcomes these limitations and significantly improves detection accuracy and robustness in complex scenarios.

2.2. Multimodal salient object detection method

In recent years, with the widespread use of multimodal sensors [40] in the field of salient object detection, deep learning-based SOD has made significant progress [41]. Zhang et al. [42] designed a two-stage multimodal fusion network that follows a “calibrate first, then fuse” principle, effectively mitigating the impact of varying-quality depth maps on the final saliency prediction. Tu et al. proposed a multi-interaction dual-decoder model that achieves accurate salient object detection by exploring and modeling the interactions among multi-level features, two modalities, and global context. Considering that dual-branch encoders typically have large model parameters, Liao et al. [43] adopted a shared encoder approach to extract complementary features

from multimodal images. Zeng et al. [44] enhanced the representation learning capability through a hybrid feature extraction network and a multi-level multimodal feature fusion module, relying on parallel attention-shift convolution to capture long-range dependencies and local information.

However, these multimodal salient object detection methods primarily focus on cross-modal feature fusion within the feature space, without considering pixel-level fusion. In practical applications, fused features can better highlight the structure of the object, enabling more effective differentiation of salient objects. Therefore, integrating multimodal feature fusion and salient object detection within a single framework can yield mutual benefits.

2.3. Underwater salient object detection

Compared to SOD, the development of USOD has been relatively short, yet there is already a substantial body of research on this topic [45]. Existing USOD methods can be broadly categorized into handcrafted feature-based methods and deep learning-based methods [46,47]. Handcrafted feature methods capture salient objects from underwater images using low-level visual features such as color, texture, and contour. Itti et al. [48] pioneered the use of local feature contrast in images to infer saliency features. Building on this foundation, researchers have proposed corresponding USOD methods [49,50]. Cui et al. [51], addressing the issues of low contrast and poor quality in underwater images, proposed an image saliency detection method based on an improved histogram equalization algorithm. Chen et al. [52] achieved underwater salient object detection by combining handcrafted features (color and intensity) with depth visual features. Islam et al. [53] guided saliency visual attention modeling by formulating bottom-up and top-down strategies within an encoder-decoder architecture.

With the rise of deep learning, various improved network architectures have been developed to address the USOD task. Li et al. [54] achieved salient object segmentation in complex underwater environments through multiple interactive feature enhancement modules and cascade decoding modules. Liu and Yu [55] pre-trained their model on a combination of terrestrial and underwater datasets and introduced a channel-spatial attention mechanism with a residual optimization module to achieve better object boundaries. To cope with complex underwater environments and the camouflaging nature of marine animals, Chen et al. [56] introduced a random style adaptation module and a twin structure to reduce the impact of underwater degradation, combined with receptive field blocks and attention fusion blocks to extract and fuse multi-level features for highlighting the object. Fu et al. addressed complex underwater imaging conditions and marine animal camouflage characteristics through a novel data augmentation strategy and a fusion-based twin deep neural network. Zhang et al. [2] enhanced feature learning in underwater images by introducing a dual structure within the SAM paradigm and adopted a multi-level coupled prompt strategy to guide underwater prior information, thereby enhancing the connection between feature learning and pixels in underwater images. Despite the significant successes of these methods, some key limitations remain, such as reduced robustness in handling blurred regions and a lack of exploration into underwater multimodal information, overlooking the potential of multimodal data to enhance underwater salient object detection.

Existing USOD methods have made certain progress in underwater salient object detection but still face several challenges. For instance, handcrafted feature methods struggle to handle low contrast and poor image quality, deep learning approaches lack robustness when dealing with blurry regions, and there is limited exploitation of multimodal information, failing to fully leverage the multi-source data in underwater environments to improve detection accuracy. In contrast, our IF-USOD method enhances salient object detection in complex underwater environments through multimodal information fusion and interactive feature enhancement, effectively utilizing the complementarity between RGB and depth information. This results in significantly improved robustness and accuracy.

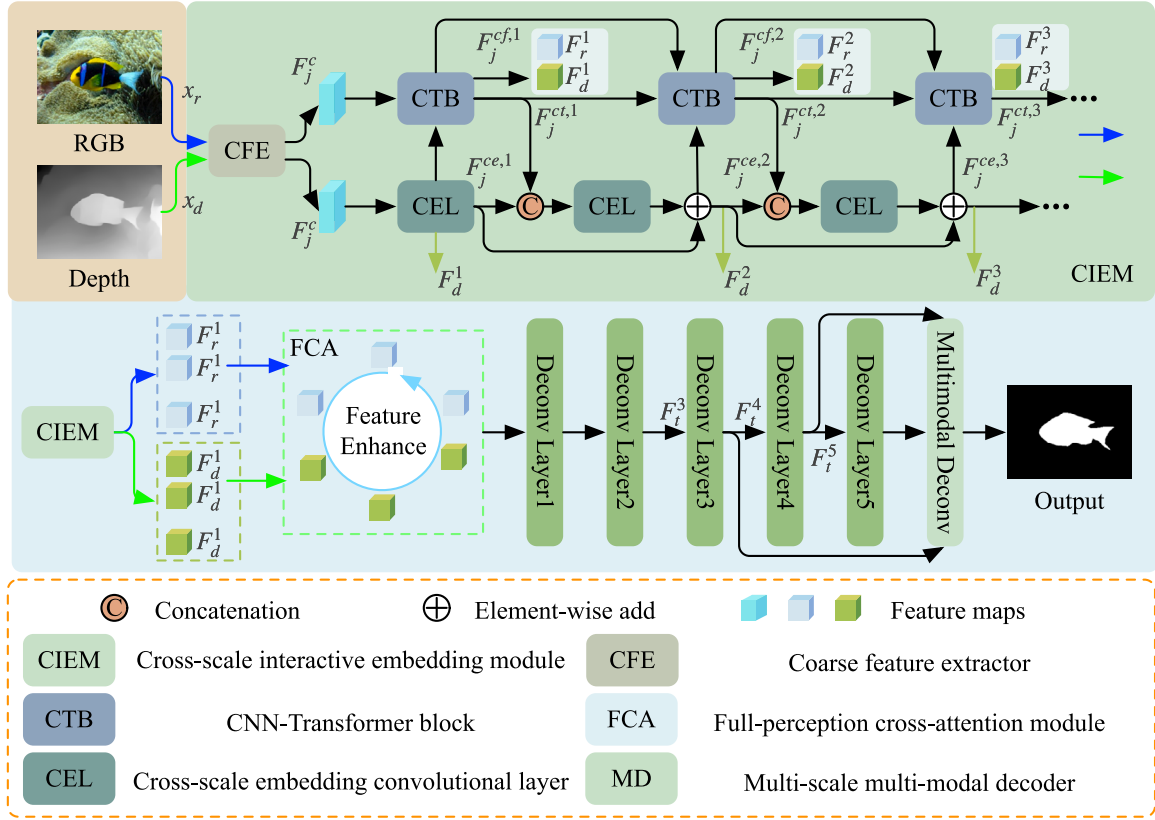


Fig. 2. Overview of the proposed interactive feature enhancement architecture for underwater salient object detection called IF-USOD. x_r and x_d are RGB and Depth image pairs. We first leverage the strengths of both CNN and Transformer architectures to extract multimodal features, followed by weighted fusion of these features based on a cross-attention mechanism. Finally, we aggregate features from different levels across multiple scales to achieve a more robust feature representation.

3. The proposed method

The proposed interactive fusion enhancement paradigm for underwater salient object detection using combined RGB and depth images is illustrated in Fig. 2. This paradigm consists of two stages: (1) the multimodal feature fusion stage, and (2) the underwater salient object detection stage. In the proposed paradigm, multimodal feature fusion facilitates top-down underwater salient object detection. Specifically, multimodal feature interaction and fusion are first achieved through the cross-scale interactive embedding module. The fused features are then fed into the full-perception cross-attention module to further enhance the object features. Finally, a multi-scale multimodal decoder is employed to perform underwater salient object detection.

3.1. Cross-scale interactive embedding module

As illustrated in Fig. 2, the CIEM primarily consists of cross-scale embedded convolutional layers (CEL) and CNN-Transformer Blocks (CTB). Preceding the CIEM module is a Coarse Feature Extractor (CFE), composed of two convolutional layers and a Leaky ReLU activation function, which is embedded to capture rich multi-channel feature maps, facilitating subsequent modules in feature extraction. We interactively exchange feature information between different levels of CTB and CEL. Specifically, features captured by CEL are input into the CTB, and features extracted by the CTB are transmitted to the next CEL. Additionally, we employ skip connections to enable information transfer between CEL and CTB at different levels.

First, we capture coarse features from the input images (x_r, x_d) using the CFE to support subsequent feature extraction by the CTB and CEL. The output features of the CFE can be represented as:

$$F_j^c = f_{cfe}(c_j), \quad j = r, d, \quad (1)$$

where, f_{cfe} denotes the feature extraction operation of the CFE, and $j = r, d$ represents the input images x_r and x_d . The extracted features F_j^c are then fed into the CTB and CEL. Unlike the ViT architecture, where input images are converted into patches to obtain local tokens, we use feature vectors formed by the features at the same spatial location across different channels of the multi-channel feature map output by the CFE as the local tokens for the Transformer in the CTB.

As shown in Fig. 3(a), the network structure of the CTB consists of a layer normalization (LN), a long-short distance attention module (LSDA), a multi-layer perceptron (MLP), and a convolutional feature representation block (CFRB). The LSDA involves both short distance attention (SDA) and long distance attention (LDA), which alternate across different blocks. Following the classic vision Transformer design, residual connections are employed within each block.

For the first CTB, i.e., $i = 1$, the inputs are F_j^c and $F_j^{ce,1}$, where $F_j^{ce,1}$ is the output of the first CEL. For the i th CTB ($i \geq 2$), let $F_j^{ct,i-1}$ be the output of the $(i-1)$ -th CTB. It is important to note that the i th CTB receives two inputs from the previous CTB. $F_j^{ct,i-1}$ is input into the LN, while $F_j^{cf,i-1}$ is input into the CFRB. Here, $F_j^{cf,i-1}$ is the output feature of the CFRB in the previous CTB. Therefore, the output of the first skip connection in the CTB is:

$$\bar{F}_j^{ct,i} = f_{ln,l\text{sd}\alpha/s\alpha}(F_j^{ct,i-1}) + F_j^{ct,i-1}, \quad (2)$$

where, $f_{ln,l\text{sd}\alpha/s\alpha}$ represents the feature extraction based on LN and SDA. The attention mechanism is divided into two parts: short distance attention and long distance attention. As shown in Fig. 3(c), for SDA, all adjacent features within a $G \times G$ grid are grouped together. In the LDA illustrated in Fig. 3(c), the input feature size is $S \times S$, and feature sampling is performed at a fixed interval I . In this case, all red boxes belong to one feature group, while all yellow boxes constitute another

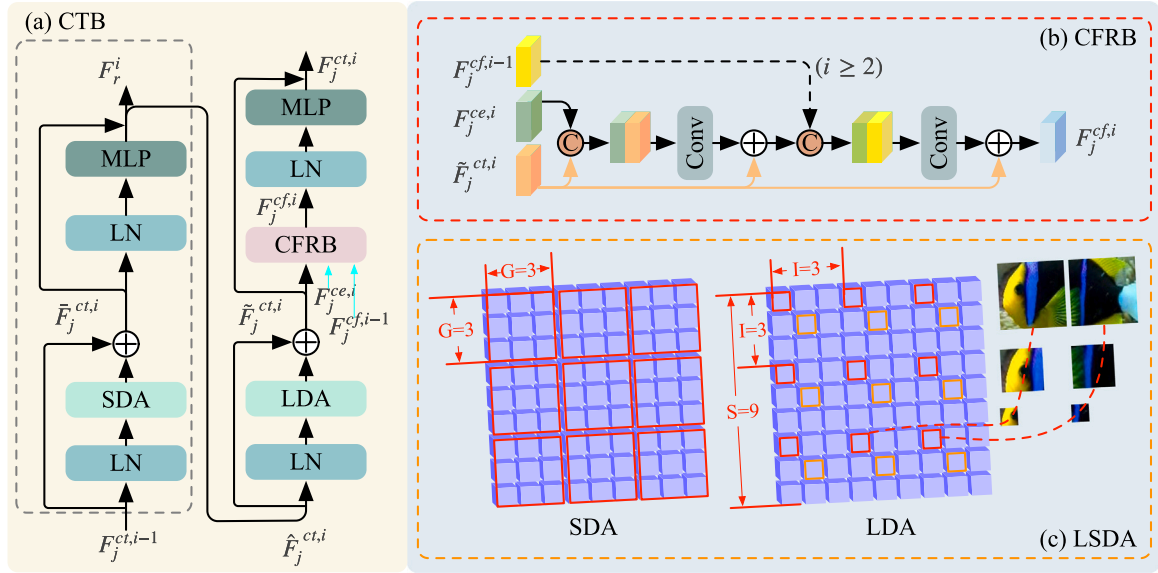


Fig. 3. The architecture of the CNN-Transformer blocks (CTB). (b) The convolutional feature representation block (CFRB). (c) The long-short distance attention module (LSDA). We incorporate convolutional blocks into the basic Transformer architecture to better integrate Transformer and convolutional features. Additionally, we utilize short-distance attention (SDA) and long-distance attention (LDA) to capture short-range and long-range attention features, respectively.

feature group. After grouping the embedded features, both SDA and LDA utilize standard self-attention mechanisms. It is worth noting that embedding large-scale adjacent patches provides sufficient context to link different features, making long-range cross-scale attention easier to compute and more meaningful [57].

The output feature of the first part in the i th CTB, denoted as $\hat{F}_j^{ct,i}$, can be expressed as:

$$\hat{F}_j^{ct,i} = f_{ln,mlp} \left(F_j^{cf,i-1} \right) + \hat{F}_j^{ct,i-1}, \quad (3)$$

where, $f_{ln,mlp}$ represents the feature extraction based on LN and MLP. The computation process of the second part in the CTB is similar to that of the first part, starting with the calculation of the skip connection:

$$\tilde{F}_j^{ct,i-1} = f_{ln,lda/llda} \left(\hat{F}_j^{ct,i-1} \right) + \hat{F}_j^{ct,i-1}, \quad (4)$$

where, $f_{ln,lda/llda}$ represents the feature extraction based on LN and LDA.

As shown in Fig. 3(b), the CFRB primarily consists of feature concatenation, convolutional layers, and skip connections. We embed the CFRB into the Transformer layer to integrate the advantages of CNNs and Transformers. The CFRB has three inputs: (1) $F_j^{cf,i-1}$, (2) the output feature $F_j^{cf,i-1}$ from the CFRB in the $(i-1)$ th CTB, and (3) the output feature $F_j^{ce,i}$ from the i th CEL. Thus, the output of the i th CFRB, denoted as $F_j^{cf,i}$, can be expressed as:

$$F_j^{cf,i} = f_{cf,rb} \left(\tilde{F}_j^{ct,i}, F_j^{cf,i-1}, F_j^{ce,i} \right), \quad (5)$$

where, $f_{cf,rb}$ denotes the feature extraction operation based on the CFRB. Therefore, the final output feature of the i th CTB, denoted as $F_j^{ct,i}$, can be expressed as:

$$F_j^{ct,i} = f_{ln,mlp} \left(F_j^{cf,i} \right) + \tilde{F}_j^{ct,i}. \quad (6)$$

The architecture of the CEL used in the CIEM is illustrated in Fig. 4. The tokens generated by the CEL can serve as inputs for the CTB at each stage. Taking the first CEL as an example, the input is the original image. We perform patch sampling using four different kernel sizes, with the stride kept consistent across all four kernels to ensure the number of generated embedded features remains the same. As shown in Fig. 4, each patch shares the same sampling center

but varies in scale. Finally, these four patches are concatenated and embedded into a single feature. The CEL enriches the representation of the input features by performing multi-scale sampling of the image with different-sized convolutional kernels and fusing this multi-scale information into an embedded representation. This approach helps the model capture important features in the image across different scales.

Specifically, the sampling and embedding processes are accomplished through four convolutional layers. Taking the D -dimensional ($D = 96$) feature shown in Fig. 4 as an example, we use convolutional kernels of sizes 4×4 , 8×8 , 16×16 , 32×32 to extract feature maps at corresponding scales, each with a stride of 4. Each feature map is processed into multiple patches, and then the patches from corresponding positions at four different scales are embedded and converted into a unified embedded representation. The output of the CEL, denoted as $F_j^{ce,i}$, can be expressed as:

$$F_j^{ce,i} = \text{Concat} \left(\text{Embed} \left(\text{Conv}_{K_i}(I) \right) \right), i = 1, 2, 3, 4, \quad (7)$$

where, $I \in \mathbb{R}^{H \times W \times C}$ represents the input image, Conv_{K_i} denotes the convolution operation with a kernel size of K , $\text{Embed}(\cdot)$ represents the embedding of feature map patches into vector representations, and $\text{Concat}(\cdot)$ indicates the concatenation of multiple embedded vectors. It is important to note that we assign a lower dimension to large kernels and a higher dimension to small kernels, thereby significantly reducing computational costs.

In the CIEM module, we facilitate the interaction of CNN features and Transformer features at different levels, effectively achieving complementary strengths in feature extraction between CNNs and Transformers. This enhances the representation capability of the captured features for image objects.

3.2. Full-perception cross-attention module

Considering that shallow features typically contain richer texture and edge details, while deep features encompass more advanced semantic information, directly fusing features at corresponding levels could inevitably result in the loss of some valuable information. To address this issue, we designed a fully perceptive feature fusion mechanism. This mechanism leverages an attention module to interactively fuse

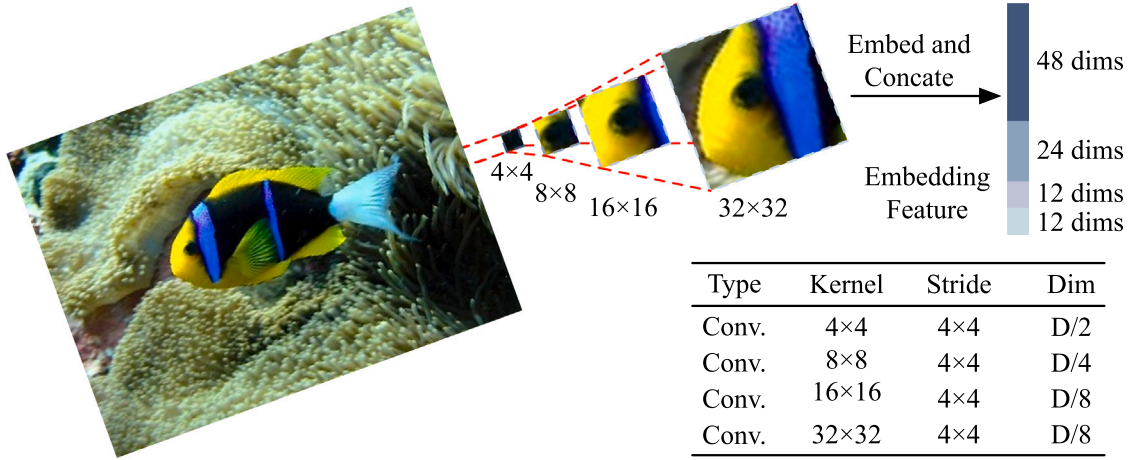


Fig. 4. The architecture of the Cross-Scale Embedded Convolutional Layers (CEL). We use four different convolution kernels (i.e., 4×4 , 8×8 , 16×16 , 32×32) to upsample the input image, where the same stride is used in the upsampling process. D represents the size of the embedded features.

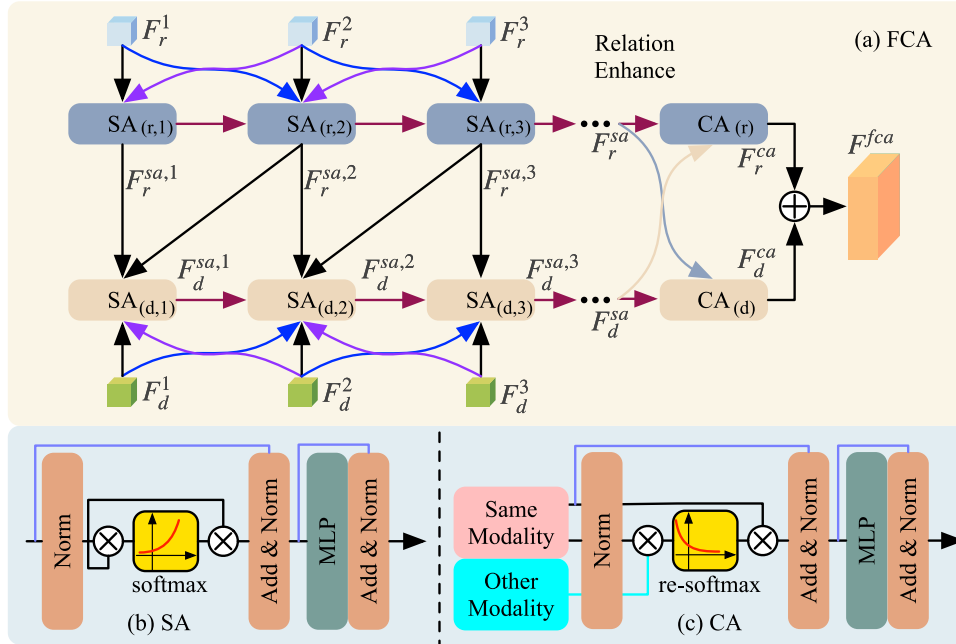


Fig. 5. The architecture of the Full-perception cross-attention module (FCA). (b) The self-attention (SA). (c) The cross-attention (CA). We adopt a cross-attention architecture to further enhance the complementarity between multimodal features while capturing more critical feature representations.

features across different levels, thereby further enhancing weak edge and texture features.

The architecture of the FCA module is illustrated in Fig. 5. We extract features from two modalities using two branches with different parameters. The features from each modality are first fed into a self-attention (SA) module to enhance internal correlations. The enhanced features are then used as inputs to the cross-attention (CA) block. The SA follows the standard Transformer architecture, which includes a self-attention module, while the CA represents a cross-attention mechanism focusing on unrelated information. For the first SA, the inputs for SA^1_r are F_r^1 and F_r^2 , while the inputs for SA^1_d are F_d^1 , F_d^2 , $F_r^{sa,1}$ and $F_r^{sa,2}$. For SA^i ($i \geq 2$), the output of the CTB, F_j^i ($i = 2, 3, \dots, n$), is used as the input for SA^i_r . The output of the SA can be computed as:

$$\begin{cases} F_r^{sa,i} = f_r^{sa,i} \left(F_r^{i-1}, F_r^i, F_r^{i+1}, f_r^{sa,i-1} \right), \\ F_d^{sa,i} = f_d^{sa,i} \left(F_d^{i-1}, F_d^i, F_d^{i+1}, f_d^{sa,i-1}, f_r^{sa,i}, f_r^{sa,i+1} \right), \end{cases} \quad (8)$$

where, $F_r^{sa,i}$ represents the SA output for the visible light image, $F_d^{sa,i}$ represents the SA output for the depth image, $f_r^{sa,i}$ is the feature-modulated SA for the visible light image, and $f_d^{sa,i}$ is the feature-modulated SA for the depth image. The calculation formula for the SA module is given by:

$$\begin{cases} [Q_j, K_j, V_j] = x_j^{sa} U_{qkv}, \\ x_j^{sa} = x_j^{sa} + \text{norm} \left(\text{softmax} \left(\frac{Q_j K_j^T}{\sqrt{d}} \right) V_j \right), \quad \text{s.t. } U_{qkv} \in \mathbb{R}^{d \times 3d}, \\ x_j^{sa} = x_j^{sa} + \text{MLP} \left(\text{norm} \left(x_j^{sa} \right) \right), \end{cases} \quad (9)$$

where, x_j^{sa} represents the input to the SA, Q_j , K_j and V_j denote different representations of the input, d is the dimensionality of the input vector, and U_{qkv} is a transformation matrix that can be learned through a fully connected layer. $\text{norm}(\cdot)$ represents the linear normalization operation, and $\text{MLP}(\cdot)$ denotes the multi-layer perceptron. The SA features of

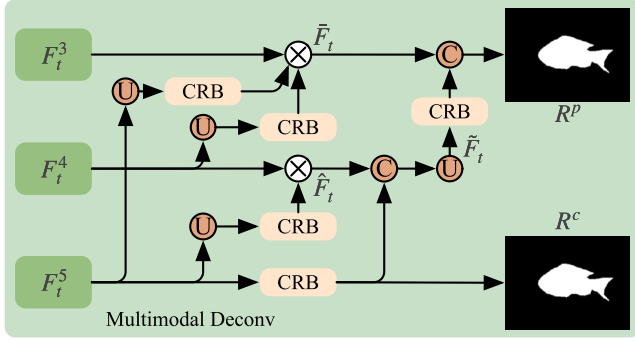


Fig. 6. The architecture of the multi-scale multimodal decoder (MD). R^c is a coarse saliency map. R^p a fine saliency map. CRB denotes Convolution + ReLU activation + Batch Normalization, U represents upsampling, F_t^i refers to features from different levels of the decoder, $\otimes$ denotes element-wise multiplication, and C indicates concatenation. $\tilde{F}_t$, $\hat{F}_t$, $\tilde{F}_t$ represents intermediate generated features.

the visible light image and depth image are then fed into the CA. The output of the CA can be computed as:

$$\begin{cases} F_r^{ca} = f_r^{ca}(F_r^{sa}, F_d^{sa}), \\ F_d^{ca} = f_d^{ca}(F_r^{sa}, F_d^{sa}), \end{cases} \quad (10)$$

where, F_r^{ca} represents the CA output for the visible light image, F_d^{ca} represents the CA output for the depth image, f_r^{ca} is the feature-modulated CA for the visible light image, and f_d^{ca} is the feature-modulated CA for the depth image. Finally, the CA outputs F_r^{ca} and F_d^{ca} are concatenated to form the output of the FCA module, denoted as F^{fca} . The calculation formula for the CA module is given by:

$$\begin{cases} [Q_j, K_j, V_j] = [x_j^{ca}, x_j^{ca}, x_j^{ca}]U_{qkv}, \\ x_j^{ca} = x_j^{ca} + \text{norm}\left(\text{softmax}\left(\frac{Q_j K_j^T}{\sqrt{d}}\right)V_j\right), \quad s.t. \ U_{qkv} \in \mathbb{R}^{3d \times 3d}, \\ x_j^{ca} = x_j^{ca} + \text{MLP}\left(\text{norm}(x_j^{ca})\right), \end{cases} \quad (11)$$

where, j and $\hat{j}$ represent the different modalities in the CA. The main difference between SA and CA lies in the activation function used after the matrix multiplication. To enhance the complementary information between different modalities, the reverse softmax (re-softmax) function is used, which better focuses on the unrelated information between the modalities. The activation function curve is shown in Fig. 5(b). The re-softmax calculation formula can be expressed as:

$$\text{re-softmax}(X) = \text{softmax}(-X). \quad (12)$$

3.3. Multi-scale multi-modal decoder

After learning the fused features, we introduce a multi-scale multimodal decoder to predict the underwater object saliency map. Unlike the traditional symmetric encoder-decoder architecture, we employ a convolutional decoder with a multi-level feature fusion strategy to decode the fused features into the saliency map. This decoder consists of six stages. The first five stages share the same structure, with each stage comprising three convolutional layers, batch normalization layers, and ReLU activation functions. As illustrated in Fig. 6, the final decoder stage involves inputting the decoded features F_t^3 , F_t^4 and F_t^5 into the multi-scale multimodal decoder module.

In the MD module, we utilize a context module and cascaded CBR (Convolution-Batch Normalization-Activation) layers to predict the saliency map. Specifically, we predict two saliency maps simultaneously: a coarse saliency map R^c and a fine saliency map R^p . The coarse saliency map provides approximate location and shape information of

the object, while the fine saliency map further refines and optimizes the object's boundaries and details. The output of the MD module can be expressed as:

$$\begin{cases} R^c = \text{CRB}(F_t^5), \\ R^p = \text{Concat}(\tilde{F}_t, \text{CRB}(\tilde{F}_t)), \\ \tilde{F}_t = F_t^3 \otimes \text{CRB}(UP(F_t^4)) \otimes \text{CBR}(UP(F_t^5)), \\ \tilde{F}_t = UP(\text{Concat}(\hat{F}_t, \text{CRB}(F_t^5))), \\ \hat{F}_t = F_t^4 \otimes \text{CRB}(UP(F_t^5)), \end{cases} \quad (13)$$

where, CRB denotes Convolution + ReLU activation + Batch Normalization, F_t^i refers to features from different levels of the decoder. $\tilde{F}_t$, $\hat{F}_t$, $\tilde{F}_t$ represents intermediate generated features. $\otimes$ represents pixel-wise multiplication, and UP refers to upsampling. Through this multimodal decoder structure, we can effectively combine and utilize different modality features, thereby significantly improving the accuracy and robustness of underwater object saliency detection.

The combination of coarse-grained and fine-grained saliency maps is designed to fully exploit feature information across different scales, thereby enhancing the overall performance of the model. Coarse saliency maps provide global contextual information, capturing the approximate location and overall outline of the target. This global information helps reduce the search area for the target and offers an initial focus point for the model, particularly in complex scenes or those with significant background noise. On the other hand, fine saliency maps focus on capturing local details and boundary information of the target, which is crucial for accurately delineating the target's boundaries and shape, especially in scenarios requiring distinction between subtle differences or similar objects. The combination of both allows the model to first quickly focus on the general region of the target through the coarse saliency map, followed by further refinement and optimization of the target's boundaries and detailed features using the fine saliency map. This multi-scale feature fusion improves detection accuracy while ensuring greater robustness when handling complex scenes.

3.4. Training strategy

To enhance the feature capturing capability of the proposed network, we adopted a two-stage training strategy. First, an encoder-decoder network was constructed for each modality to reconstruct the input images. Subsequently, the entire network was trained using multimodal inputs and the proposed total loss function.

In the first stage, our goal is to train the encoders to extract rich features, laying the foundation for the subsequent USOD task. Due to the differences between RGB images and depth images, we use different encoder-decoder networks to reconstruct the input images. As shown in Fig. 7, the RGB branch utilizes a Transformer architecture, while the depth branch employs a convolutional architecture. Additionally, we use skip connections to preserve both deep and shallow features in the network. During the training process of the first stage, we employ Structural Similarity Index (SSIM) loss to train the network.

In the first stage, the RGB and depth image encoder-decoder networks are independently trained, allowing the networks to focus on extracting rich features from each single modality. Given the significant feature differences between RGB and depth images, using Transformer and convolutional architectures respectively helps to better capture the characteristic information of each modality. By training the network with the SSIM loss function, the perceptual quality consistency of the reconstructed images is ensured, effectively enhancing the feature extraction capability of the encoders. This phased training approach enables the encoders to extract features with higher precision and improved robustness in subsequent tasks.

The second stage of training involves the standard end-to-end training process, as illustrated in Fig. 2. In this stage, the entire CIEM, FCA,

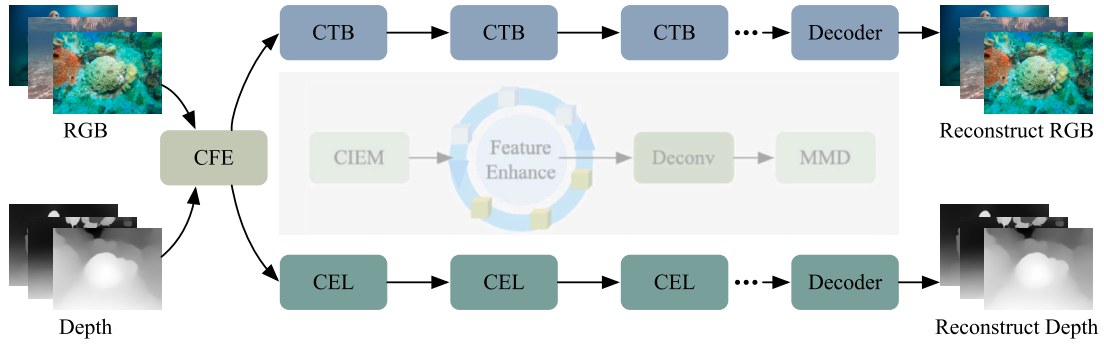


Fig. 7. Two-stage training strategy. We first pre-train the network by reconstructing images, and then train the full model architecture based on this.

and MD modules are trained together. During the second stage, we utilize Weighted Binary Cross-Entropy (wBCE) loss, Weighted IoU (wIoU) loss, Dice loss, and SSIM loss to supervise the network's training. The specific configuration of the loss functions is detailed in Section 3.5. It is important to note that the encoder parameters from the first stage of training are not fixed; instead, they are used as pre-trained models.

In the second stage, the entire network is trained with multimodal inputs, during which the CIEM, FCA, and MD modules are fully optimized. The use of multimodal inputs allows the network to facilitate the interaction and fusion of information between different modalities, thereby enhancing the model's adaptability to complex scenarios. The combined use of multiple loss functions, including wBCE, wIoU, Dice, and SSIM, enables the network to be optimized under multiple objectives. This joint supervision by multiple loss functions helps the network improve classification accuracy while reducing false positives and false negatives, thereby enhancing the fineness of image segmentation and structural consistency.

Using the encoder parameters obtained from the first stage as a pretrained model provides a strong initialization for the second stage of training. This approach not only accelerates the convergence speed of the network but also effectively mitigates the instability that can arise from random initialization. Additionally, since the pretrained model already possesses strong feature extraction capabilities, the network can more rapidly adjust and optimize the feature fusion between different modalities during multimodal joint training, leading to superior segmentation results. The two-stage training strategy, through staged feature extraction and fusion optimization, enables the proposed network to fully leverage its capabilities in feature capture and fusion when tackling complex USOD tasks. This not only enhances the training efficiency and convergence of the network but also achieves high-precision and high-robustness image segmentation in the joint training of multimodal data.

3.5. Loss function

We use Weighted Binary Cross-Entropy loss and Weighted IoU loss to supervise the network training. In saliency detection, the object regions (positive samples) are typically much smaller than the background regions (negative samples). The wBCE effectively balances the imbalance between positive and negative samples through weighting, thereby improving the detection performance for object regions. The weighting mechanism allows the model to focus more on hard-to-classify samples, enhancing the model's robustness and generalization ability. IoU is a common evaluation metric in saliency detection, and the wIoU loss directly optimizes the IoU, enabling the model to better improve the overlap between the predicted and ground truth results during training. The wIoU loss effectively suppresses false positives and false negatives during the optimization process, reducing misdetections and missed detections, thus improving detection accuracy.

Given the output coarse saliency map R^c and fine saliency map R^p , the coarse loss is calculated as follows:

$$\mathcal{L}_{coarse} = \mathcal{L}_{bce}^w(R^c, GT) + \mathcal{L}_{iou}^w(R^c, GT), \quad (14)$$

The fine loss is calculated as:

$$\mathcal{L}_{precise} = \mathcal{L}_{bce}^w(R^p, GT) + \mathcal{L}_{iou}^w(R^p, GT), \quad (15)$$

SOD loss can be expressed as:

$$\mathcal{L}_{sod} = \mathcal{L}_{coarse} + \mathcal{L}_{precise}. \quad (16)$$

The combined use of wBCE and wIoU loss functions allows for complementary benefits in sample balancing and overlap optimization, significantly enhancing the overall performance of saliency detection. The joint supervision of the coarse and fine saliency maps enables the model to be optimized at different levels, resulting in more refined and accurate saliency detection outcomes.

Additionally, we employ Dice and SSIM losses to further supervise the model training process. Dice loss assesses the similarity between two samples, while SSIM loss supervises the structural information within the image, enabling the model to better learn the boundary information of salient objects. Therefore, the total loss function can be expressed as:

$$\mathcal{L}_{total} = \mathcal{L}_{sod} + \mathcal{L}_{dice} + \mathcal{L}_{ssim}. \quad (17)$$

4. Experiments

We evaluate the model on underwater salient object detection datasets such as USOD10K and COD10K. These datasets contain both RGB images and corresponding depth maps, enabling multimodal fusion. Additionally, we conduct generalization studies on the USOD and SOC datasets to verify the model's generalization performance. The model is quantitatively analyzed using classical evaluation metrics such as S-measure, F-measure, E-measure, and MAE. Visual comparisons of different methods are also provided to demonstrate the superiority of the IF-USOD model in underwater salient object detection.

4.1. Datasets

USOD10K is a large-scale dataset specifically designed for underwater salient object detection [4]. This dataset consists of 10,255 underwater images, featuring 12 different underwater scenes and 20 different types of salient objects. USOD10K includes not only RGB images but also the corresponding depth maps and boundary annotations for each image. The aim of USOD10K is to improve the detection

and segmentation of salient objects in underwater scenes by providing a diverse and complex set of images that reflect real underwater conditions.

The COD10K dataset is a high-quality dataset specifically designed for camouflaged object detection [58]. It contains over 10,000 high-resolution images, covering a wide range of scenes and object types. Among these, underwater detection objects are a crucial component of the camouflaged object detection task. The underwater objects in the COD10K dataset mainly include various marine organisms, such as fish, turtles, sea snakes, starfish, and sea urchins. These organisms often possess protective coloration or camouflage abilities, allowing them to evade predators or prey in their natural environments. In addition to natural organisms, the dataset may also include some artificial objects, such as shipwrecks, diving equipment, and other debris, which often blend with the surrounding background in underwater environments, further increasing the difficulty of detection. The camouflaged object detection task is inherently challenging because camouflaged objects are typically highly similar to their background in terms of color, texture, and shape, making them difficult for traditional image segmentation and detection algorithms to identify effectively.

The USOD dataset [53] is specifically designed for salient object detection in underwater environments. The USOD dataset comprises thousands of underwater images from various scenes, including oceans, lakes, and rivers, representing a wide range of environments. The dataset provides medium to high-resolution images, typically ranging from 300×300 to 600×600 pixels, which retain sufficient detail to allow algorithms to extract meaningful features of the salient objects. Salient objects in the dataset include a variety of underwater entities, such as fish, marine animals, coral reefs, submerged objects, and plants.

The SOC dataset is a high-quality dataset specifically designed for studying visual search and saliency detection [59]. SOC is primarily used for research on salient object detection tasks across different scenes and backgrounds. The SOC dataset comprises 3569 images from over 80 common categories, including 2412 training images and 1159 testing images. These images feature various complex backgrounds and multiple types of salient objects, aiming to provide a standardized benchmark for saliency detection algorithms in the field of computer vision. In this paper, we use the SOC dataset to evaluate the generalization performance of our model.

Note that when depth images are not provided in the dataset, we use the DPT method [60] to generate the corresponding depth images to achieve multimodal training.

4.2. Implementation details

We conducted experiments on the USOD10K and COD10k dataset. During the training phase, each pair of RGB and depth images was randomly cropped into 140×140 image patches. We used the Adam optimizer for parameter updates during training, with a batch size set to 4. The initial learning rate was set to 1×10^{-4} . Throughout the network training, the learning rate gradually decayed following a cosine schedule down to 1×10^{-6} . For fair comparison, all deep learning-based methods were retrained on the USOD10K dataset using default settings, and the evaluation results were assessed using the same evaluation code.

4.3. Evaluation metrics

In this work, we use the following four metrics to evaluate the performance of the model. Structural Similarity Measure (S_s) [61] assesses the structural similarity between the predicted image and the ground truth. S_s typically combines luminance, contrast, and structural information to evaluate the quality of the segmentation results, providing a better reflection of the perceived quality of the image.

$$S_s = \alpha S_o + (1 - \alpha) S_r, \quad (18)$$

Table 1

Performance comparison on USOD10k. The best and second results are in red and blue, respectively.

Methods	$S_s \uparrow$	$E_s \uparrow$	$F_\beta^\omega \uparrow$	$MAE \downarrow$
F3Net	0.9140	0.9599	0.9171	0.0251
DANet	0.9006	0.9449	0.8934	0.0279
CSNet	0.8595	0.9178	0.8642	0.0548
MFNet	0.8425	0.9146	0.8193	0.0512
BTSNet	0.9093	0.9542	0.9104	0.0291
PSGLoss	0.8640	0.9078	0.8508	0.0417
SwinFusion	0.9067	0.9528	0.9136	0.0315
TC-USOD	0.9215	0.9683	0.9236	0.0201
CMX	0.9083	0.9552	0.9108	0.0336
BDLFusion	0.8659	0.9047	0.8626	0.0485
HiDANet	0.8901	0.9429	0.9035	0.0522
Dual-SAM	0.9238	0.9684	0.9311	0.0185
CrossFuse	0.9121	0.9585	0.9176	0.0288
Our	0.9240	0.9688	0.9312	0.0180

where, S_o and S_r represent the object-aware and region-aware structural similarity, respectively, with α set to 0.5.

The Weighted F-measure (F_β^ω) [62] combines precision and recall metrics, introducing a weighting factor β to emphasize the balance between precision and recall. F_β^ω effectively addresses the issue of imbalanced datasets by adjusting the different weights of precision and recall, making it more flexible in practical applications.

$$F_\beta^\omega = (1 + \beta^2) \frac{P^\omega R^\omega}{\beta^2 P^\omega + R^\omega}, \quad (19)$$

where, β represents the effectiveness of the detection, P^ω denotes the weighted precision, and R^ω denotes the weighted recall.

The Adaptive E-measure (E_s) [63] takes into account both pixel-level matching and image-level statistical information. E_s specifically focuses on the similarity between the segmented image and the ground truth, including the alignment of overall structure and local details.

$$E_s = \frac{1}{w \times h} \sum_{x=1}^w \sum_{y=1}^h \Phi(x, y), \quad (20)$$

where, h and w represent the height and width of the segmented result image, respectively. Φ is the enhanced alignment matrix.

The Mean Absolute Error (MAE) measures the average of the absolute errors between the predicted results and the ground truth labels, representing the difference between the predicted and actual values. MAE provides a straightforward and intuitive reflection of the deviation in the prediction results.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (21)$$

where, y_i represents the ground truth, and $\hat{y}_i$ represents the predicted value.

4.4. Comparisons with the state-of-the-arts

In this section, we compare the proposed method with other state-of-the-art approaches. The results of quantitative evaluation and qualitative analysis clearly demonstrate the advantages of our proposed method. We compared our approach against 13 advanced methods, including deep SOD methods: F3Net [15], CSNet [13], MFNet [6], PSGLoss [10]; deep RGB-D SOD methods: DANet [7], BTSNet [17], CMX [14], HiDANet [12]; USOD methods: TC-USOD [4], Dual-SAM [2], SwinFusion [23]; and deep RGB-T methods: BDLFusion [20], CrossFuse [21].

Quantitative Evaluation. Table 1 summarizes the performance of 13 salient object detection methods on the USOD10K dataset. It is clear from the table that our method outperforms all competitors across all quantitative evaluation metrics, demonstrating higher segmentation accuracy and lower error rates. TC-USOD and Dual-SAM also perform well, particularly in the E_s and F_β^ω metrics, coming close to our method,

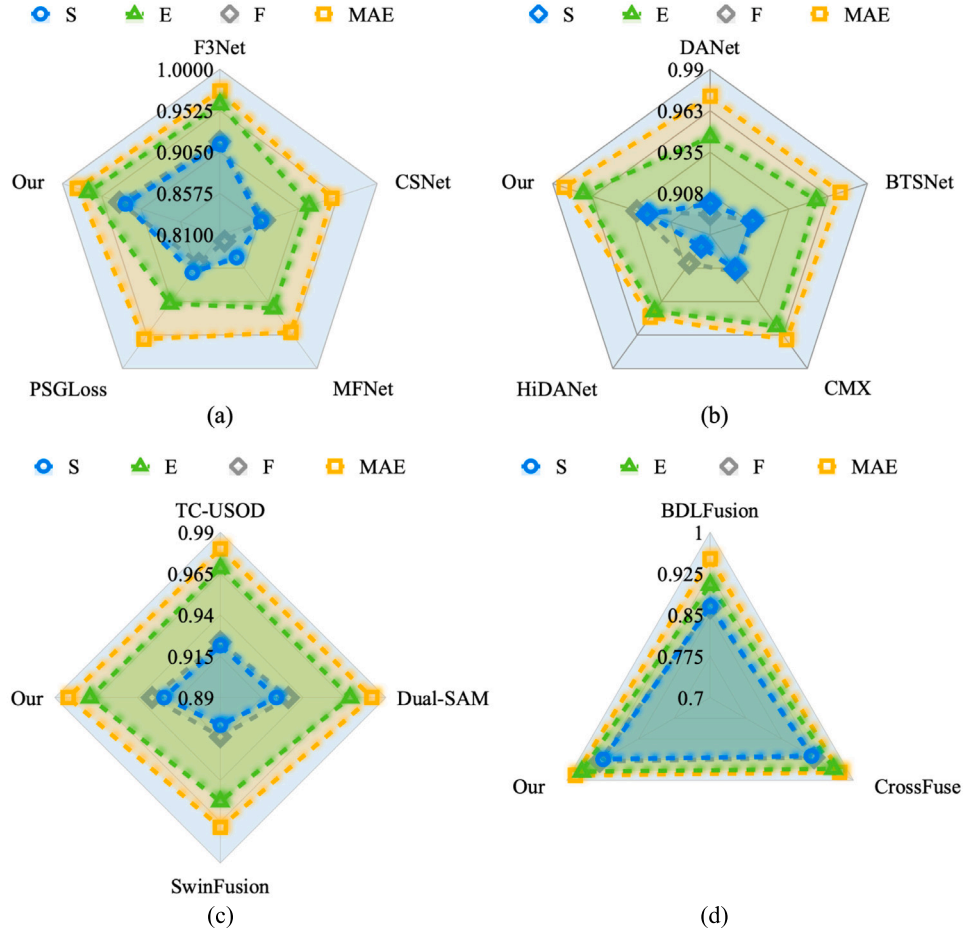


Fig. 8. Visualization of quantitative evaluation results. (a) The comparison with classical multimodal fusion methods. (b) The comparison with dual-stream fusion architectures. (c) The compares our method with state-of-the-art approaches. (d) The comparison with attention mechanisms.

but showing a slight gap in MAE . F3Net, BTSNet, and CrossFuse perform well on the S and E metrics but fall short in MAE compared to our method. CSNet and MFNet score relatively low across all metrics, especially in MAE , indicating poorer performance in handling errors. Specifically, in the S_α evaluation, our method outperforms TC-USOD and DualSOD by 0.0025 and 0.0002, respectively. It surpasses CSNet by 0.0645, indicating a stronger ability to maintain overall structural consistency in the image. In the E_ξ evaluation, our method exceeds TC-USOD's 0.9683 and Dual-SAM's 0.9684, showing a slight advantage in edge alignment consistency. Compared to MFNet, our method improves by 0.1263, further proving a significant enhancement in detail capture and edge precision. In the F_β^ω evaluation, our method outperforms the lower-scoring SwinFusion by 0.0176 in the F_β^ω , demonstrating its superior ability to balance precision and recall.

A comprehensive analysis of Table 1 reveals that MFNet and PSGLoss exhibit certain shortcomings in handling image complexity and details, which makes it difficult for the models to accurately align edges in specific scenarios, thereby affecting overall segmentation performance. HiDANet and BDLFusion may struggle with extracting and fusing features from different information sources, particularly in complex scenes or with multimodal inputs, where the models might find it challenging to maintain high segmentation accuracy, leading to a decline in overall performance. SwinFusion and CrossFuse achieve better results across different evaluation metrics, mainly due to their effective handling of multi-scale information, optimized feature fusion strategies, and precise detail capture. SwinFusion's strength lies in its Transformer architecture and multi-scale feature processing capability, while CrossFuse ensures robustness and high precision in complex scenes through

cross-modal information fusion and feature alignment strategies. Our method excels in all four evaluation metrics, particularly in MAE and S_α . This success is largely attributed to the cross-scale interactive embedding module and fully perceptive cross-attention module, which not only effectively capture multimodal features but also facilitate interactive learning between these features, thereby fully leveraging the complementary advantages of multimodal features.

Fig. 8 further illustrates the evaluation results of different algorithms on the USOD10k dataset. We utilize radar charts to better illustrate the quantitative evaluation results. Fig. 8(a) presents a comparison with classical multimodal fusion methods. F3Net enhances the model's ability to capture details through multi-level feature fusion and attention mechanisms. However, in the S_α evaluation, F3Net shows slightly lower structural similarity due to its insufficiency in integrating global information. Fig. 8(b) shows the comparison with dual-stream fusion architectures. DANet employs a dual attention mechanism that effectively captures global contextual information, thereby improving its performance in the S_α and E_ξ metrics. Fig. 8(c) primarily compares our method with state-of-the-art approaches. TC-USOD enhances the capture of image structure and edge information by introducing a context-aware mechanism, leading to better results in the E_ξ and F_β^ω evaluations. Finally, Fig. 8(d) evaluates the performance differences across various attention mechanisms. SwinFusion, CrossFuse, and our method exhibit unique advantages in multi-scale information processing and feature fusion, enabling them to achieve excellent performance in global structure consistency, edge alignment, and error control.

Qualitative Analysis. Fig. 9 presents the qualitative comparison results of SOD and USOD algorithms on the USOD10k dataset. It

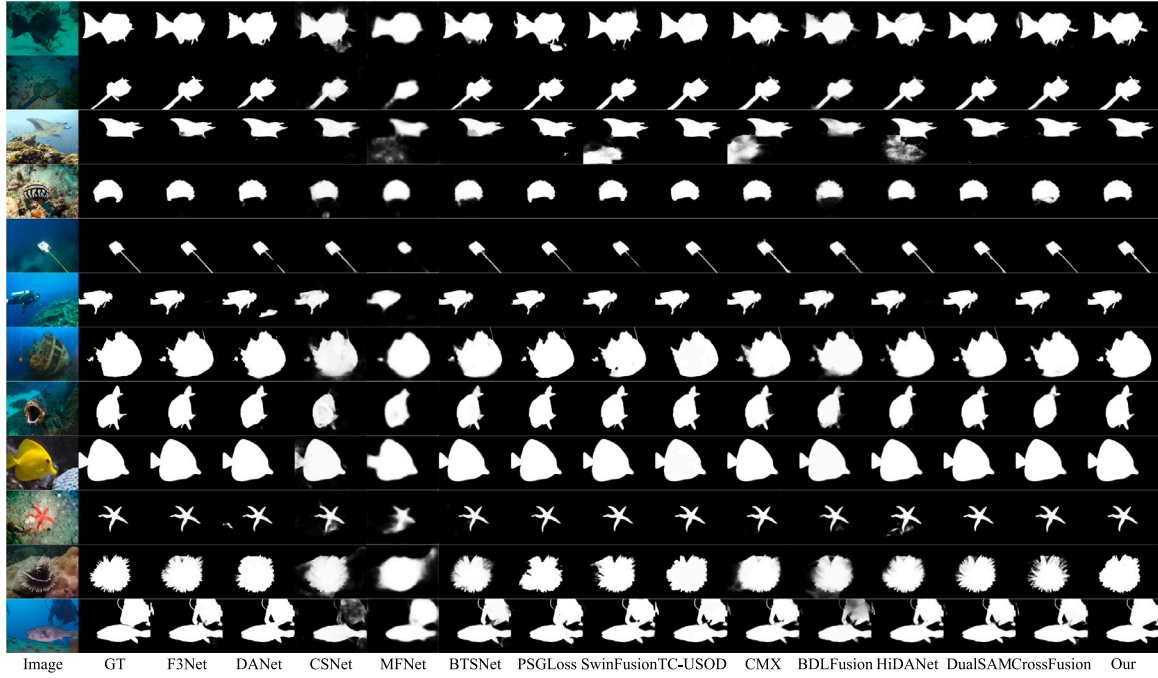


Fig. 9. Qualitative analysis results on the USOD10K dataset.

is evident that our proposed method achieves superior accuracy, almost perfectly matching the ground truth (GT), and excels in edge, detail, and shape preservation. Whether dealing with objects in complex backgrounds or those with rich edge details (such as the spiny organism in the second-to-last row), our model accurately segments the object area while avoiding noticeable noise or errors. SwinFusion and TC-USOD also demonstrate high accuracy in most cases, with clear segmentation edges that closely align with the GT. Particularly when handling complex backgrounds (such as images of divers and underwater creatures), these methods effectively distinguish between foreground and background, maintaining the shape integrity of the objects. While SwinFusion and TC-USOD perform consistently with good edge processing and detail preservation, they are slightly less effective than our proposed method in some extreme cases (e.g., images with significant noise or blurry areas). CrossFuse and Dual-SAM excel in processing images with relatively simple shapes and high contrast. However, when dealing with objects with complex shapes or rich details (such as starfish or objects with intricate edges), they sometimes exhibit blurred edges or detail loss. F3Net and DANet are capable of accurately identifying and separating objects during segmentation, but in certain scenarios (e.g., scenes with complex backgrounds or fuzzy object edges), the segmentation edges may become blurred or affected by noise, leading to less accurate object shapes (as seen in the objects in the third and second-to-last rows). PSGLoss, BDLFusion, and HiDANet perform poorly when handling images with complex edges or low contrast. Comparing the segmentation results of different algorithms, it is clear that the proposed model consistently demonstrates the most stable and accurate performance across various scenarios, effectively handling complex backgrounds, detail-rich scenes, and maintaining the shape integrity of objects.

As shown in Table 2, on the COD10K dataset, our model demonstrated strong competitiveness. Firstly, in the S_a evaluation, the model scored 0.8850, an improvement of 0.0483 over F3Net's 0.8367, and also surpassing SwinFusion's 0.8772. This significant improvement indicates that the model better maintains structural consistency in images when handling camouflaged objects with complex backgrounds. The FCA module plays a crucial role in balancing local details and global information by integrating shallow and deep features. In the E_z evaluation, the model scored 0.9032, showing an improvement of 0.0195 over

Table 2

Performance comparison on COD10k. The best and second results are in red and blue, respectively.

Methods	$S_a \uparrow$	$E_z \uparrow$	$F_\beta^\omega \uparrow$	$MAE \downarrow$
F3Net	0.8367	0.8837	0.8125	0.0642
DANet	0.8451	0.8853	0.8726	0.0416
CSNet	0.7928	0.8258	0.8116	0.1120
MFNet	0.7883	0.8261	0.7572	0.1206
BTSNet	0.8519	0.8937	0.8663	0.0607
PSGLoss	0.8546	0.8575	0.8418	0.0974
SwinFusion	0.8772	0.9013	0.8945	0.0311
TC-USOD	0.8475	0.9114	0.8858	0.0412
CMX	0.8217	0.8348	0.8015	0.1137
BDLFusion	0.8157	0.8246	0.7662	0.1328
HiDANet	0.8614	0.8750	0.8641	0.0533
Dual-SAM	0.8755	0.8906	0.8870	0.0587
CrossFuse	0.8721	0.9022	0.8993	0.0440
Our	0.8850	0.9032	0.9014	0.0317

F3Net and surpassing SwinFusion by 0.0019. This suggests that our model has a stronger ability to align and process complex edges. The introduction of the SSIM loss further enhances this advantage, enabling the model to better focus on edge information, thereby improving overall segmentation accuracy.

For the F_β^ω metric, our proposed model increased by 0.0889 compared to F3Net and improved by 0.0069 over SwinFusion. This result further demonstrates the model's superiority in balancing precision and recall, especially in the task of segmenting underwater objects. The Dice loss significantly enhanced the model's ability to capture salient regions, allowing it to maintain a high level of segmentation quality across different background complexities. Finally, the model achieved an MAE of 0.0317, significantly lower than F3Net's 0.0642 and SwinFusion's 0.0311. This performance indicates the model's excellent overall error control, further confirming the effective combination of the FCA module and multiple loss functions. This allows the model to effectively reduce errors when dealing with detail-rich and noisy images.

Fig. 10 presents the qualitative analysis results of various state-of-the-art models on the COD10K dataset. By comparing the performance of these models in camouflaged object detection tasks, we can better understand the strengths and weaknesses of each approach when

dealing with diverse and complex underwater scenes. From the results shown by F3Net, it is clear that while the model can detect the main regions of camouflaged objects, it struggles with boundary definition. It often produces segmented outputs with noticeable artifacts or excessive smoothing, especially in complex environments such as coral reefs or densely vegetated areas. This suggests that F3Net has limitations in maintaining fine edge details and accurately segmenting objects with complex shapes. In contrast, DANet demonstrates improved edge preservation, particularly in simpler scenes. However, in regions with complex structures or low contrast, the model fails to clearly delineate object boundaries, resulting in blurred segmentation outputs. Although DANet's dual attention mechanism aids in capturing global context, it may still fall short when handling complex textures and intricate details. CSNet and MFNet exhibit similar shortcomings, with segmentation results that often lack clarity and consistency. Both models appear to struggle with capturing fine details, frequently producing incomplete or fragmented object masks. This issue is particularly evident in images with highly camouflaged objects, where the models fail to effectively distinguish the object from the background. In comparison, SwinFusion and TC-USOD show significant improvements in segmentation accuracy, especially in preserving the overall shape and structure of camouflaged objects. SwinFusion, leveraging its Transformer-based architecture, excels in capturing global dependencies and aligning them with local details, resulting in more precise segmentation in scenarios with high background complexity. Similarly, TC-USOD benefits from its context-aware mechanism, enhancing edge alignment and object delineation.

The results shown in Fig. 10 clearly indicate that our model outperforms all other methods across all test cases. Our approach maintains high precision in both edge definition and internal structure of the objects, regardless of the background complexity or the degree of camouflage. The CIEM and FCA modules, by integrating shallow and deep features and enhancing them, significantly boost the model's performance. Our segmentation results are closer to the GT, demonstrating that our model exhibits superior robustness when dealing with various underwater environments and complex objects. The ability to accurately segment camouflaged objects in complex underwater scenes further underscores the effectiveness of the CIEM and FCA modules, as well as the importance of carefully designed loss functions in guiding the training process.

Extended Generalization. Table 3 presents the quantitative evaluation results of various methods on the USOD and SOC single-modal datasets. On the USOD dataset, our model demonstrates outstanding performance in the E_ξ , F_β^ω , and MAE metrics. Specifically, the proposed method improves the E_ξ score by 0.0085 compared to F3Net's 0.9493, significantly outperforming most of the compared methods. TC-USOD achieves the highest E_ξ score, slightly surpassing our method, which may be attributed to its unique design for edge handling and global information capture. In the F_β^ω evaluation, our model shows a slight decrease compared to F3Net. However, this minor decline might be due to the trade-offs made in optimizing other metrics, such as MAE. In terms of MAE, the proposed method excels, significantly outperforming F3Net and all other methods, indicating a notable advantage in detail preservation and error control. Although our method shows a slight decrease in the S_α metric compared to TC-USOD, overall, our model demonstrates strong comprehensive capabilities when balancing multiple evaluation metrics, particularly excelling in error control and edge alignment.

On the SOC dataset, our method also exhibits superior performance. In the S_α evaluation, our model slightly surpasses HiDANet's 0.8662 and Dual-SAM, indicating that the proposed method can better maintain structural consistency of the image in complex backgrounds. In contrast, F3Net performs relatively poorly on the SOC dataset, likely due to its difficulty in effectively distinguishing object regions from the background when dealing with complex backgrounds, leading to a decrease in overall structural similarity. In terms of E_ξ , our method's

performance is slightly lower than TC-USOD but still significantly outperforms other methods, demonstrating its capability in edge alignment and handling complex edges. Notably, in the MAE metric, our model scores significantly lower than TC-USOD, indicating better overall error control, especially in reducing errors when handling high-noise and complex scenes.

In terms of S-measure and F-measure, our approach underperforms compared to SwinFusion and TC-USOD on the SOC dataset. This could be attributed to the proposed method's limitations in capturing global structures and object boundaries in single-modal data. For E-measure and MAE metrics, the TC-USOD method demonstrates superior ability in distinguishing salient regions from the background. In contrast, our method may struggle with fine-grained salient object detection in complex environments. It is worth noting, however, that our method consistently exhibits outstanding generalization across various scenarios, accurately capturing salient objects in complex visual scenes, which underscores the efficacy of our feature extraction and fusion strategies. While TC-USOD and SwinFusion excel in certain detail-oriented aspects, our method maintains comprehensive and robust performance across multiple evaluation metrics, demonstrating strong competitiveness in handling diverse datasets.

IF-USOD consistently demonstrates strong performance on both multimodal and unimodal datasets, primarily due to its innovative multimodal information fusion and interactive feature enhancement architecture. First, by integrating RGB and depth information, the model fully leverages the complementarity between different modalities, addressing the limitations of single modalities (e.g., RGB or depth alone) in accurately capturing salient objects in complex scenes. Second, the cross-scale interactive embedding module in the IF-USOD model effectively combines the strengths of convolutional neural networks and Transformer architectures. Moreover, the fused features are re-weighted by the fully-aware cross-attention module, further enhancing cross-modal complementarity to ensure that the fused images maintain outstanding performance in downstream vision tasks. Finally, IF-USOD employs a multi-scale multimodal decoder, which integrates multi-scale, multi-source features from various levels to produce precise saliency prediction results.

4.5. Ablation study

In this work, the proposed IF-USOD employs a hybrid architecture that integrates a Transformer encoder and a convolutional encoder. To validate the effectiveness of this architecture, we used a purely convolutional encoder-decoder architecture as Baseline 1 and a purely Transformer encoder-decoder architecture as Baseline 2. These baselines allow us to assess the effectiveness of the proposed modules under these conditions.

Table 4 illustrates the impact of different module combinations on model performance. We analyzed the baseline models based on convolution and Transformer architectures and the performance improvements achieved by incrementally adding different modules. Baseline 1 (Conv) uses a traditional convolutional neural network approach. The evaluation results show that, without using a Transformer, the convolutional approach exhibits relatively weaker performance in overall structural similarity and edge alignment, with higher errors. Baseline 2 (Transformer) introduces the Transformer architecture, which significantly enhances performance. Compared to Baseline 1, its E_ξ and F_β^ω scores improved by 0.0526 and 0.0973, respectively, indicating that the Transformer has clear advantages in processing global information and capturing details. Base IF-USOD (Transformer+Conv) demonstrates that integrating convolutional operations on top of the Transformer further improves performance. The S_α and F_β^ω evaluations increased by 0.0224 and 0.0111, respectively, showing that the introduction of convolutional operations helps better capture local features, enhancing the model's overall performance. Base IF-USOD (+CTB) shows that the CTB module significantly improves the model's ability to handle

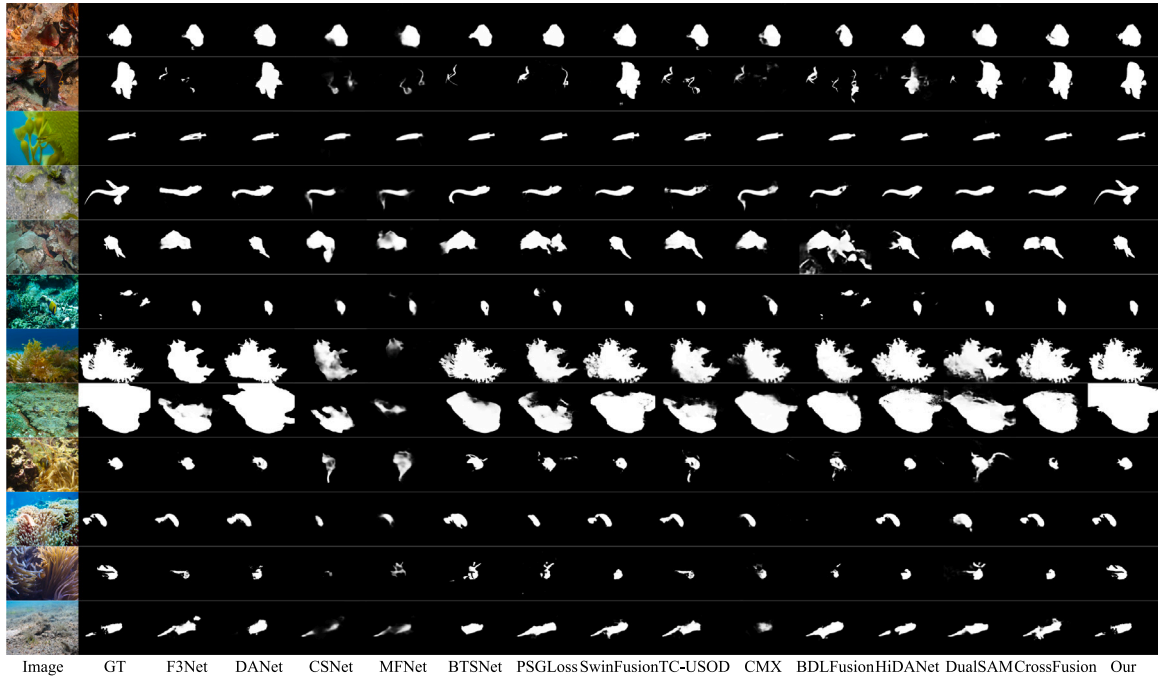


Fig. 10. Qualitative analysis results on the COD10K dataset.

Table 3

Quantitative evaluation on USOD and SOC datasets. The best and second results are in red and blue, respectively.

Methods	USOD				SOC			
	$S_a \uparrow$	$E_c \uparrow$	$F_{\beta}^w \uparrow$	$MAE \downarrow$	$S_a \uparrow$	$E_c \uparrow$	$F_{\beta}^w \uparrow$	$MAE \downarrow$
F3Net	0.9122	0.9493	0.9263	0.0367	0.7283	0.7911	0.6283	0.1042
DANet	0.8959	0.9369	0.9099	0.0410	0.6390	0.6835	0.4268	0.1772
CSNet	0.8668	0.9166	0.8697	0.0759	0.6946	0.7825	0.5633	0.1162
MFNet	0.8465	0.9103	0.8485	0.0722	0.7984	0.8092	0.7440	0.1258
BTSNet	0.9007	0.9351	0.9110	0.0471	0.6937	0.7486	0.5275	0.1432
PSGLoss	0.8911	0.9366	0.9108	0.0416	0.8314	0.8428	0.8305	0.0972
SwinFusion	0.9033	0.9514	0.9133	0.0356	0.8740	0.8509	0.8867	0.0554
TC-USOD	0.9245	0.9672	0.9217	0.0218	0.8550	0.9026	0.8847	0.0558
CMX	0.9053	0.9525	0.9038	0.0441	0.7836	0.7881	0.7648	0.1568
BDLFusion	0.8614	0.8968	0.8469	0.0522	0.8015	0.8000	0.7540	0.1487
HiDANet	0.8834	0.9316	0.8950	0.0575	0.8662	0.8601	0.8539	0.0688
Dual-SAM	0.9168	0.9546	0.9237	0.0260	0.8541	0.8727	0.8672	0.0738
CrossFuse	0.8955	0.9337	0.8855	0.0352	0.8624	0.8825	0.8640	0.0544
Our	0.9134	0.9578	0.9225	0.0213	0.8672	0.8855	0.8854	0.0471

Table 4

Performance comparison of different modules.

Methods	$S_a \uparrow$	$E_c \uparrow$	$F_{\beta}^w \uparrow$	$MAE \downarrow$
Baseline 1 (Conv)	0.8123	0.8995	0.8032	0.0683
Baseline 2(Transformer)	0.8834	0.9521	0.9005	0.0436
Base IF-USOD (Transformer+Conv)	0.9058	0.9517	0.9116	0.0342
Base IF-USOD (+CTB)	0.9122	0.9534	0.9175	0.0311
Base IF-USOD (+CTB+FCA)	0.9231	0.9668	0.9303	0.0189
Base IF-USOD (+CTB+FCA+MD)	0.9240	0.9688	0.9312	0.0180

complex structures, resulting in enhanced structural similarity and edge alignment performance. Base IF-USOD (+CTB+FCA) confirms that the FCA module contributes to improving the model's ability to capture global features, thereby significantly enhancing segmentation accuracy and consistency. Base IF-USOD (+CTB+FCA+MD) demonstrates that the MD module further enhances the model's performance in complex scenarios by optimizing multi-scale information fusion, particularly excelling in detail processing and error control.

The results in Table 4 clearly indicate that the stepwise addition of different modules significantly enhances the model's segmentation

Table 5

Ablation study of CIEM module.

Methods	$S_a \uparrow$	$E_c \uparrow$	$F_{\beta}^w \uparrow$	$MAE \downarrow$
Base IF-USOD (Transformer+Conv)	0.9058	0.9517	0.9116	0.0342
Base IF-USOD (+SDA)	0.9065	0.9521	0.9138	0.0341
Base IF-USOD (+SDA+LDA)	0.9095	0.9531	0.9168	0.0322
Base IF-USOD (+SDA+LDA+CFRB)	0.9122	0.9534	0.9175	0.0311

performance. The combination of Transformer and convolutional operations establishes a strong performance foundation for the baseline model, while the gradual introduction of CTB, FCA, and MD modules brings substantial improvements in structural similarity, edge alignment, weighted F-measure, and error control.

Effectiveness of the CIEM Module. The core advantage of the CIEM module lies in its ability to efficiently capture and fuse cross-scale information. A key challenge in USOD tasks is effectively capturing both global and local information across different scales. While traditional convolutional neural networks excel in capturing local features, they often fall short in integrating global information. The CIEM module, through the combination of SDA and LDA, enables the interaction

Table 6

Base IF-USOD refers to the use of both Transformer and CNN encoders, upon which we verify the effectiveness of various modules. “w/o *” indicates the absence of the * architecture in the model.

Methods	$S_a \uparrow$	$E_\xi \uparrow$	$F_\beta^\omega \uparrow$	$MAE \downarrow$
Base IF-USOD (Transformer+Conv)	0.9058	0.9517	0.9116	0.0342
Base IF-USOD (w/o CA)	0.9158	0.9514	0.9175	0.0237
Base IF-USOD (w/o SA)	0.9156	0.9482	0.9119	0.0251
Base IF-USOD (+FCA)	0.9231	0.9668	0.9303	0.0189

and integration of features across multiple scales. This design allows the model to retain detailed information while effectively integrating global semantics, especially when dealing with complex backgrounds or highly diverse images. Moreover, the CIEM module fully leverages the strengths of both CNN and Transformer architectures. Convolutional layers are adept at extracting local spatial features, while Transformers excel in capturing global dependencies and long-range information. By alternately using CEL and CTB within the CIEM, the model can dynamically adjust feature representations across different scales. This hybrid architecture enables the model to provide fine segmentation edges in detail-rich regions while maintaining structural consistency on a global scale. It is also worth noting that convolutional operations often generate a variety of feature representations across different channels, but these features may not be fully utilized in traditional networks. The CFRB module reorganizes and reweights features across channels, allowing the model to more effectively harness this information. This feature reorganization helps reduce redundancy and enhances focus on critical features, thereby further improving the model's performance in edge alignment and detail preservation.

Table 5 presents the results of ablation studies conducted on the CIEM module, aiming to assess the impact of the CEL and CTB on model performance. By progressively incorporating these modules, their contributions to various evaluation metrics can be observed. Initially, the performance of the base model (Base IF-USOD (Transformer+Conv)) is weak. Upon introducing the SDA module, the model's performance improves, with S_a increasing by 0.0007, E_ξ by 0.0004, F_β^ω by 0.0022, and a slight decrease in MAE by 0.0001. These results indicate that the SDA module effectively enhances feature capture and edge alignment, thereby improving overall segmentation performance. Further performance gains are observed with the addition of the LDA. Specifically, S_a increases by 0.0030, E_ξ by 0.0010, F_β^ω significantly increases by 0.0030, and MAE decreases by 0.0019. The introduction of the LDA module enhances the interaction between multi-scale features, effectively improving the model's ability to retain detail and maintain global consistency. Finally, with the inclusion of the CFRB module, the model reaches its optimal performance. S_a improves from 0.9095 by 0.0027 to 0.9122. E_ξ increases by 0.0003, reaching 0.9534. F_β^ω further increases by 0.0007, reaching 0.9175. and MAE decreases by another 0.0011. The CFRB module enhances the flow of information between channels, further improving the model's overall performance, particularly in handling complex backgrounds and edge details. The CIEM module, through effective feature processing and multi-scale information fusion, significantly enhances the overall performance of the image segmentation model.

Effectiveness of the FCA Module. The outstanding performance of the FCA module is attributed to its fully perceptive fusion mechanism across different feature levels. Shallow features often contain rich texture and edge information, while deep features carry high-level semantics, both of which are crucial for segmentation tasks. Simply fusing these features at their corresponding levels might result in the loss of some critical information. The FCA module addresses this by introducing a cross-attention mechanism, allowing for fine-tuned control of feature fusion at each level, ensuring that the model fully leverages the information from every layer, thereby enhancing overall segmentation performance. Specifically, the combination of CA

Table 7

Ablation study of loss function.

Methods	$S_a \uparrow$	$E_\xi \uparrow$	$F_\beta^\omega \uparrow$	$MAE \downarrow$
IF-USOD (precise)	0.9121	0.9487	0.9125	0.0236
IF-USOD (precise+coarse)	0.9155	0.9515	0.9135	0.0210
IF-USOD (precise+coarse+dice)	0.9159	0.9511	0.9240	0.0202
IF-USOD (precise+coarse+dice+ssim)	0.9240	0.9688	0.9312	0.0180

and SA enables the FCA module to dynamically adjust the interactions between features, allowing the model to deliver more accurate segmentation results, particularly when dealing with complex backgrounds and edges.

Table 6 presents the ablation study results of the FCA module, where key components of the FCA were progressively removed or retained to evaluate their impact on model performance. The FCA module is designed to enhance the representation of weak edges and texture features by interactively fusing features at different levels, thereby improving the model's performance in complex scenarios. The evaluation results in the table indicate that the base model performs well in handling global semantic information and capturing local details, but there is still room for improvement in optimizing weak edges and texture features. Upon removing the CA module, the model's performance shows slight variations: S_a improves by 0.0100, E_ξ decreases slightly by 0.0003, F_β^ω increases by 0.0059, and MAE significantly decreases by 0.0105.

The removal of the CA module leads to subtle changes in how the model handles cross-channel features, likely due to the absence of inter-channel information weighting. This results in the model relying more on spatial information fusion, leading to enhanced detail processing but a slight reduction in global feature integration. After removing the SA module, S_a remains close to the results observed after removing the CA module, but E_ξ decreases by 0.0035 and F_β^ω increases slightly by 0.0003. The removal of the SA module leads to a slight decline in edge alignment and global consistency, suggesting that SA plays a critical role in reinforcing edge feature alignment and improving the model's ability to capture details.

Effectiveness of the Loss Functions. The evaluation results in **Table 7** reflect the critical role of different loss functions in the model training process. The wBCE and wIoU loss functions serve as the foundation, equipping the model with the ability to balance sample equilibrium and overlap optimization. The joint supervision of coarse and fine saliency maps allows the model to be optimized at different scales, enabling better handling of complex image scenarios. The Dice loss further optimizes the similarity between the predicted and object regions, significantly enhancing the model's accuracy in saliency detection. The SSIM loss, by supervising the structural information of the image, ensures that the model not only excels in numerical metrics but also maintains a high degree of structural consistency in visual appearance.

Table 7 demonstrates the impact of different loss function combinations on the performance of the IF-USOD model. By progressively introducing various loss functions, the specific contribution of each loss function to the improvement of model performance becomes evident. The experimental results indicate that using only precise saliency map supervision already allows the model to achieve certain performance levels in structural similarity and edge alignment, but it still shows limitations in detail capture and overall error control. When coarse saliency map supervision is added to precise saliency map supervision, the model's performance improves further. Specifically, S_a increases by 0.0034, E_ξ by 0.0028, F_β^ω by 0.0010, and MAE significantly decreases by 0.0026. The introduction of the coarse saliency map enables the model to capture global information on a larger scale, thereby enhancing its adaptability to complex scenes and making object region detection more robust. Adding the Dice loss on top of the previous configuration results in further performance enhancement. The Dice loss, by directly optimizing the similarity between the object and

predicted regions, effectively improves the model's ability to capture salient regions, particularly in terms of detail processing and regional similarity. Finally, the SSIM loss optimizes the structural similarity of the image, ensuring that the model excels in capturing details while maintaining structural consistency. This is especially beneficial for improving segmentation accuracy when dealing with complex edge information.

5. Conclusion

In this work, we propose a novel USOD architecture named IF-USOD. This architecture includes a dual structure with Transformer and CNN components to better extract features from underwater images. Additionally, to better represent the correlation between features, we propose using self-attention and cross-attention mechanisms to reaggregate the features. In our proposed method, multimodal feature fusion is more inclined towards underwater salient object features, aligning with the requirements of the USOD task. We also employ multiple loss functions to enforce the inclusion of more salient features and detailed information in the fused features. Extensive experiments demonstrate that our proposed method achieves state-of-the-art performance in USOD tasks, and ablation studies validate the effectiveness of the proposed modules. In future work, we will further explore dynamic adjustments in feature fusion to meet the practical demands of USOD in extreme conditions.

CRedit authorship contribution statement

Genji Yuan: Writing – original draft, Software, Data curation, Conceptualization. **Jintao Song:** Validation, Resources, Formal analysis. **Jinjiang Li:** Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

No potential competing interests exist. There is no an undisclosed relationship that may pose a competing interest. There is no an undisclosed funding source they may pose a competing interest. Moreover, The author corresponding author, have full access and responsibility for the accuracy of data analysis, manuscript preparation and decision to submit the manuscript for publication.

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Data availability

Data will be made available on request.

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